XIAMEN UNIVERSITY MALAYSIA

EEE313
Semiconductor Physics and Device Processes
Introduction

Dr. Mohammadmahdi Ariannejad
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Office: A1-418
mohammadmahdi.ariannejad@xmu.edu.my

Welcome...

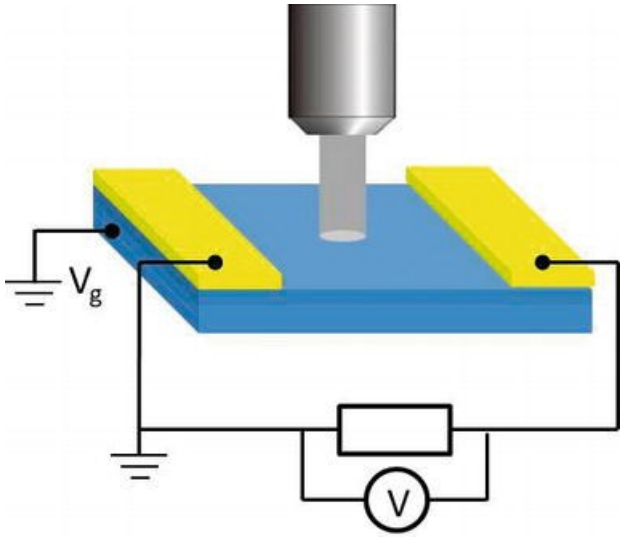
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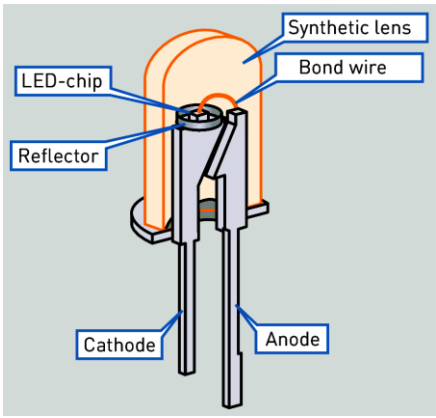




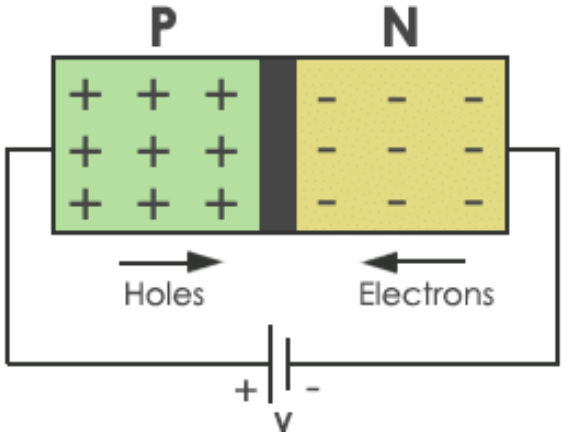








This course provides a detailed introduction to the principal classes of semiconductor devices in **modern microelectronics and photonics**. The following topics will be discussed; **charge carrier statistics** and **transport, luminescence**, **photoconductivity**, **p-n junctions**, **metal- semiconductor junctions**, **diodes**, **field-effect transistor (MOSFET), bipolar junction transistor (BJT)**, **light-emitting diodes (LED), laser diodes (LD), photodiodes, and photovoltaic solar cells**.









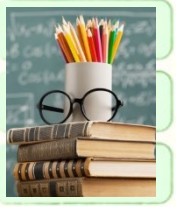
Introduction





- Lecturer: Dr. Mohammadmahdi Ariannejad (Dr. Mahdi)
  - Email: mohammadmahdi.ariannejad@xmu.edu.my
- Your Lecturer's background
  - PhD in photonics engineering (UM)
  - Master in microelectronics engineering (UKM)
  - Bachelor in Electronics Engineering (Electrical)

Consultation hours

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
1						
2						
3						
4	4 PM - 6 PM					
5				5 PM - 7 PM		
6						
7						
8						



Monday:

4:00 PM to 6:00 PM

Thursday:

5:00 PM to 7:00 PM

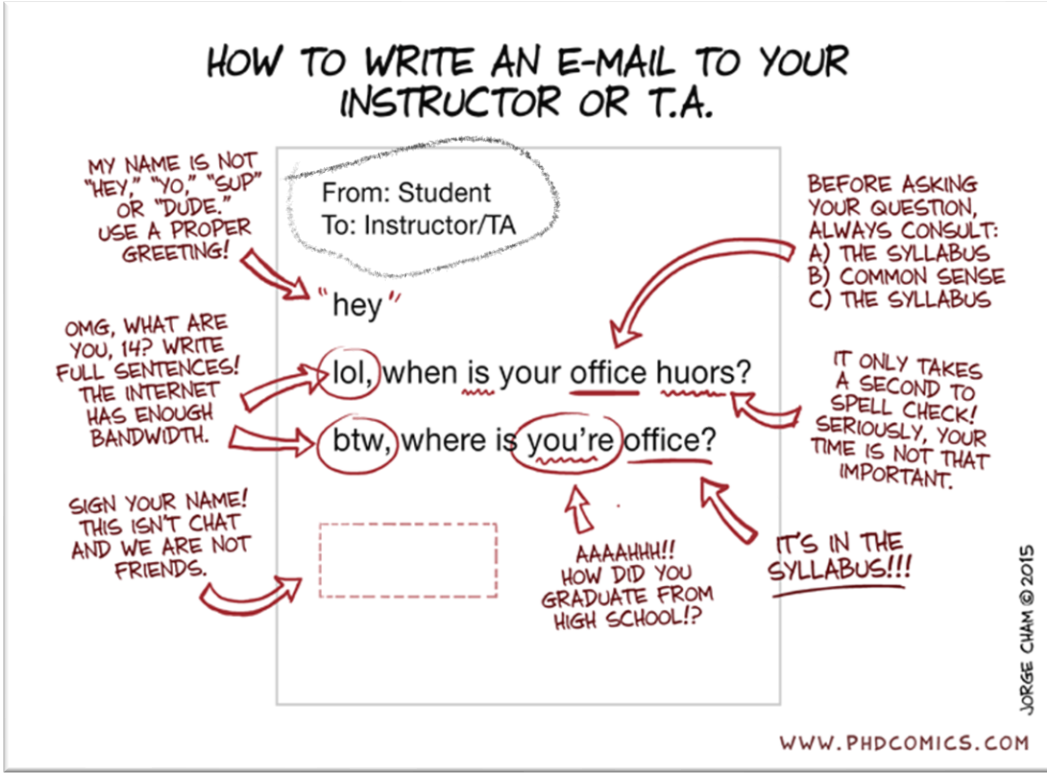


- Office location: A1-418
- Please email first!
- Send email for consulting in any other time slot

Staff: Whom do I contact?



- Lecturer
  - For help with teaching and learning materials, e.g., lecture contents or tutorial questions
  - Mid-semester exam, final exam
- Administration – Office of Academic Affairs
  - Enrolment problems, Attendance issues….



Source: http://phdcomics.com/comics/archive.php?comicid=1795

Course Learning Outcome (CLO)

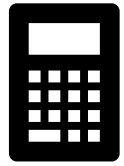
1. **Explain** the physical concepts underlying the operation of **semiconductor devices** and able to **draw and interpret energy band diagrams** (C2).

2. **Analyze** the behavior of a **p-n junction**, **BJT**, and **FET including** device physics, device operation, and device characteristics and their performances (C4).

3. **Design** semiconductor devices with the physical concepts underlying the operation of advanced semiconductor devices including **solar cells and LEDs** (C6).

Bloom's Taxonomy

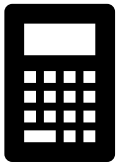
	Cognitive (thinking, knowledge)
Lower Order	C1 Remembering
	C2 Understanding
Intermediate Order	C3 Applying
	C4 Analysing
Higher Order	C5 Evaluating
	C6 Creating



Program Learning Outcome (PLO)



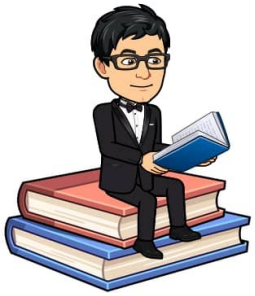
- Engineering knowledge
- Problem Analysis



- Design/development of solutions



What I am expecting to learn in here?

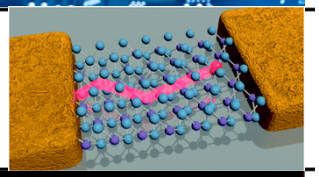


Course Content Outline

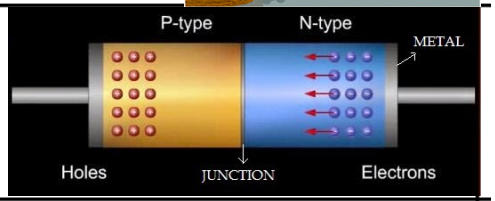
1. Semiconductor Concepts
1.1. Molecular orbitals
1.2. Lattice structure of solid
1.3. Energy bandsbandgap
1.4. Density of states
1.5. Fermi-Dirac function



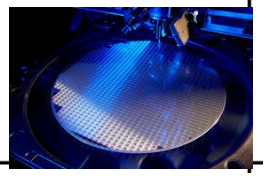
2. Carrier Transport
2.1. Carrier drift and diffusion, generation and recombination
2.2. Carrier injection
2.3. Surface states and recombination

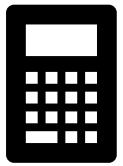


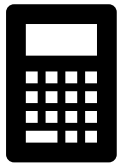
3. P-N junction
3.1. Forward and reverse bias
3.2. Junction breakdown,
3.3. PN junction diode
3.4. Generation–recombination currents

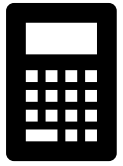


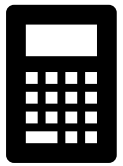
4. Fundamental and fabrication processes of semiconductor devices
4.1. Metal–Semiconductor and Semiconductor Heterojunctions
4.2. Fundamentals of the Metal–Oxide–Semiconductor Field-Effect Transistor
4.3. LEDs and Lasers
4.4 Photodetectors and Solar Cells











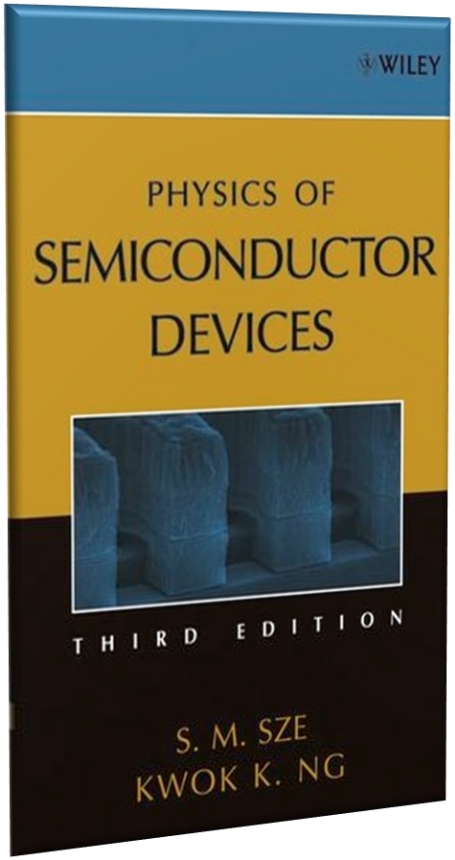


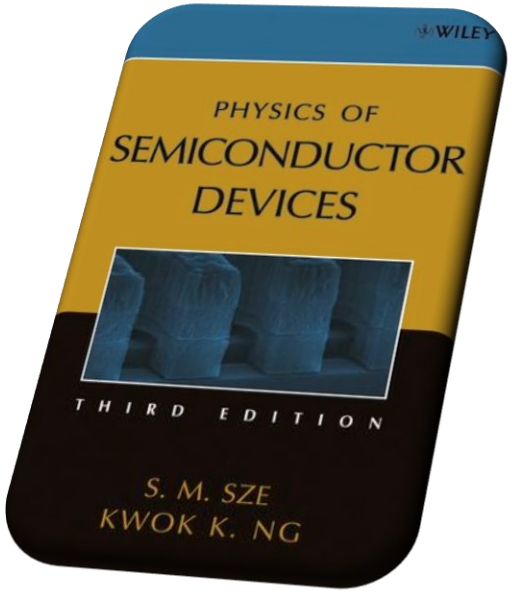
EEE313- Timetable

Class type	Venue	Time	
Lecture	B1#209	Wednesday : 01:00 PM- 03:00 PM	
Tutorial	A1#G02	Group 1	Friday: 09:00 AM- 10:00 AM
		Group 2	Friday: 10:00 AM- 11:00 AM

Reference Book







- S. M. Sze, Y. Li, K. K. Ng, Physics of Semiconductor Devices, 4th Ed. Wiley 2019.
- S. M. Sze, M.-K. Lee Semiconductor Devices: Physics and Technology, 3rd Edition, Wiley 2012.

Prerequisites

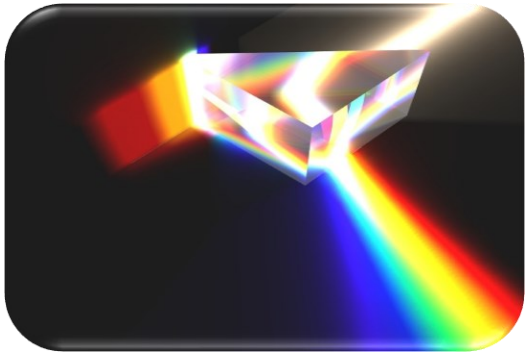


- Engineering Physics II









Course activities

- Lectures
  - Lecture outlines available on Moodle
  - Download these to your device, or print as handouts and bring to lectures as note taking aids

- Tutorials
  - Questions will be available on Moodle
  - Please prepare for the tuts before coming to the class
  - Active participation – **Be brave – we're friendly!**

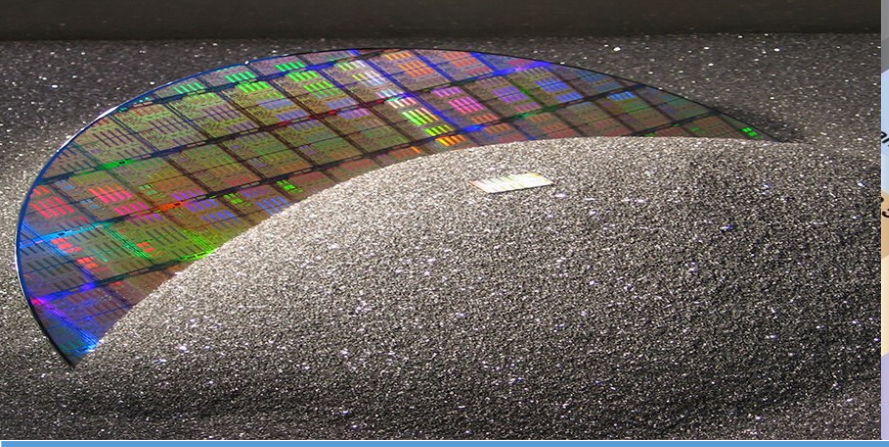
- Assessments







EEE313	Assessment	Ratio
	Assignment	30%
	Quiz	20%
	Final exam	50%



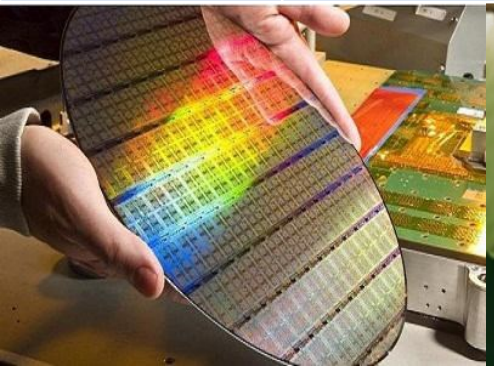


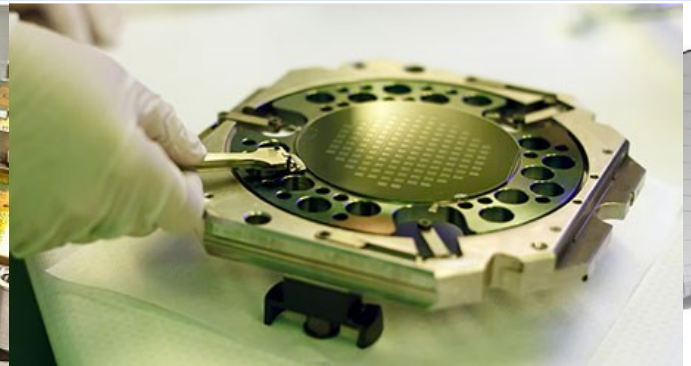
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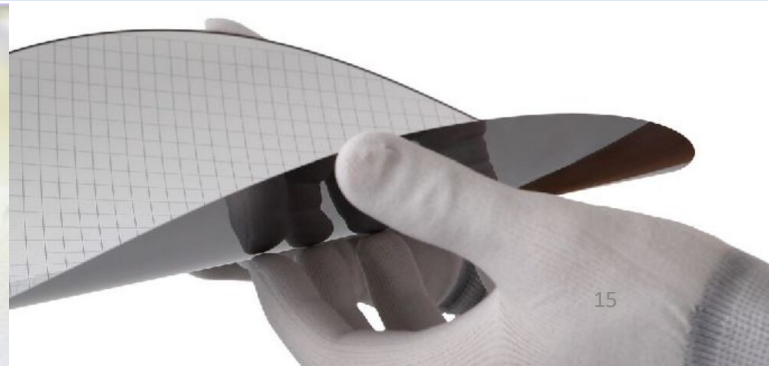
EEE313

Semiconductor Physics and Device Process

Part I	Semiconductor material properties
Part II	Fundamental semiconductor devices











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How to learn

1. Do not miss class. Pay attention and participate in the class. Check Moodle announcement frequently.

2. Do the assignments by yourself. Do not copy and also do not let other copy your work. You are encouraged to discuss among yourselves and with me.

3. Think frequently and solve problem (reinforcement)

What we expect from you?

- Interactive
- Motivated to learn
- Independent
- Punctual



Grading System

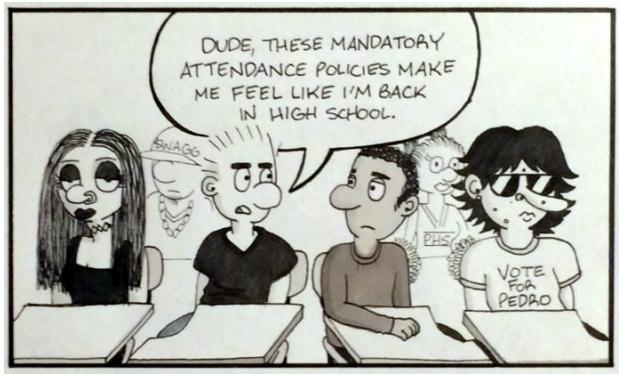
- Grading Scales and grades applicable to ALL intakes effective from Academic Session 2017/09:

Marks	Grade	Grade Points	Description
85 – 100	A	4.00	Distinction
80 – 84	A-	3.70	Distinction
75 – 79	B+	3.30	Good
70 – 74	B	3.00	Good
65 – 69	B-	2.70	Good
60 – 64	C+	2.30	Pass
55 – 59	C	2.00	Pass
50 - 54	C-	1.70	Conditional Pass
45 – 49	D+	1.30	Fail
40 - 44	D	1.00	Fail
≤ 39	F	0.00	Fail

University policy on attendance

- Important: You are required to have an 80% attendance to be able to sit for the final exam

- Not acceptable reasons for below 80% attendance
  - If I attend the rest of the class then my attendance will be 80%
  - My attendance is almost 80% (But if it is below 80% then it does not meet the requirements)
  - ….



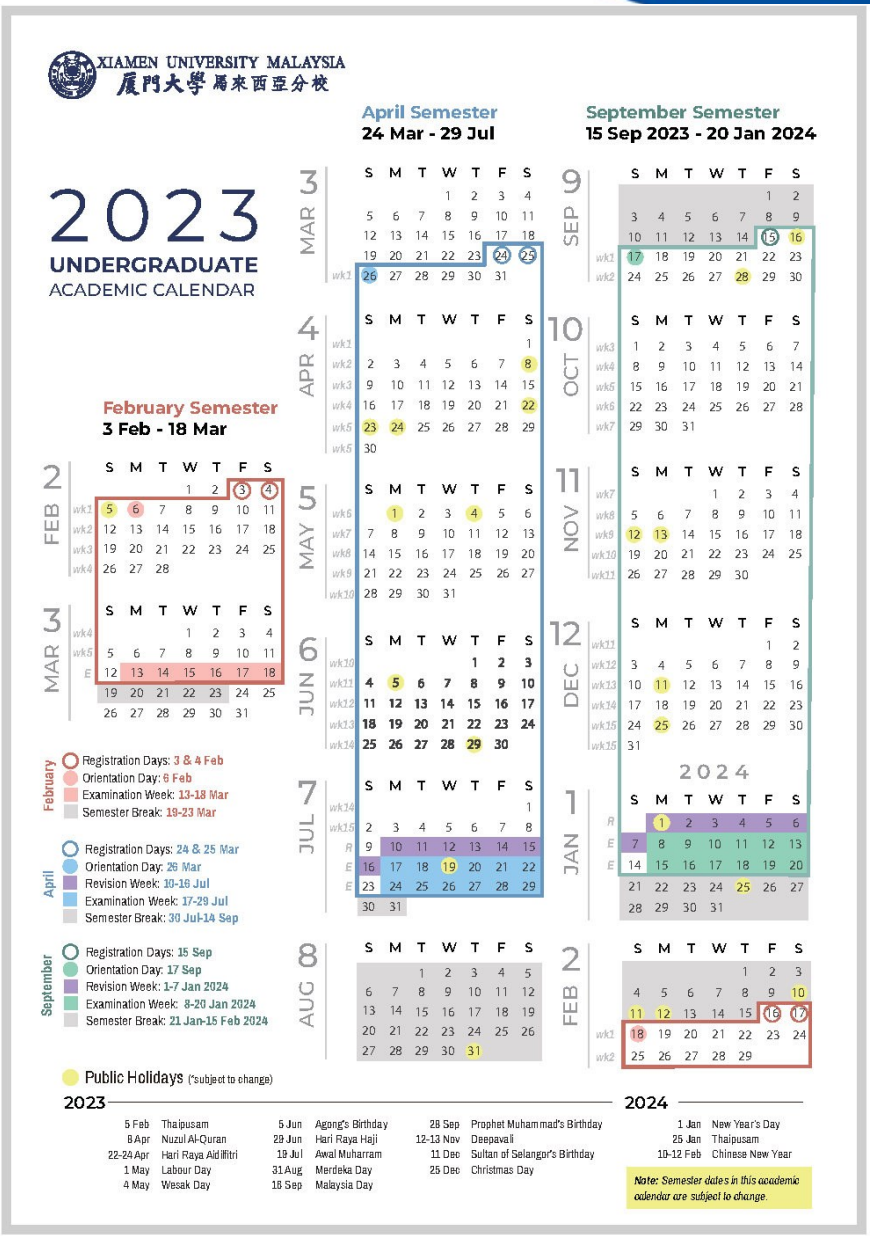


Holidays and feedbacks!

- Feel free to provide any suggestions to improve the delivery of this course

No Public Holiday





Course webpage

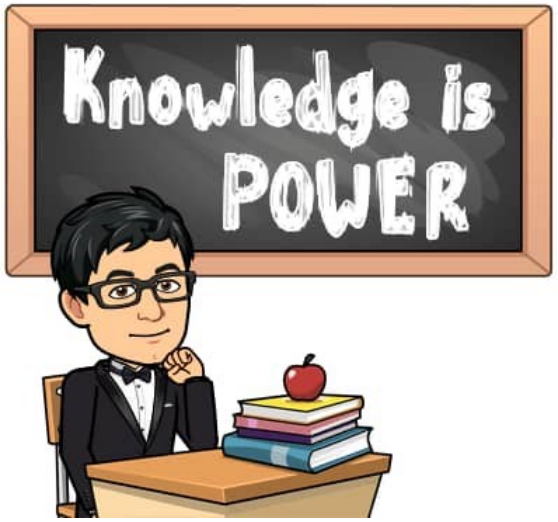
- https://l.xmu.edu.my/my/

- Search for: **EEE313 Semiconductor Physics and Device Processes 2023/04 Mohammadmahdi Ariannejad**

- Enrolment Key: siliconsemicon202304



XMUM Moodle



Announcements

Reference book

S. M. Sze, Y. Li, K. K. Ng, Physics of Semiconductor Devices, 4th Ed. Wiley 2019

Assignment Template

Template of Assignment Cover

- Please **use** this file as your **cover page** for the assignments.

Exam/Quiz Templates

EEE313 - Template of Aswer Book Cover_202204

- Please **use** this file as your **cover page** for the exam or Quiz

Template of Student Answer Sheet - 202204

Please **use this file** to write your answers **neatly**.
Remember you need to send it to me in **PDF format**! other formats won't be accepted.

Week 1



Teaching plan (Tentative)

Week 1 to week 3: semiconductor material properties	Semiconductor Concepts-molecular orbitals
	Lattice structure of solid
	Energy bands and bandgap
	Density of states, Fermi-Dirac function
Week 3 to week 6: Carrier Transport (Week 5 Assignment 1)	Carrier drift and diffusion, generation and recombination
	Carrier injection
	Surface states and recombination
Week 7 to week 10 (Quiz 1): **P-N junction and PN junction diode** Forward and reverse bias, junction breakdown, generation–recombination currents Schottky barriers. (Week 12 Assignment 2)	
Week 10 to week 15 (Quiz 2): Fundamental of semiconductor devices Metal–Semiconductor and Semiconductor Heterojunctions, Fundamentals of the Metal–Oxide–Semiconductor Field-Effect Transistor, Optical Devices. LEDs and Lasers, photodetectors and Solar Cells.	



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EEE313

Semiconductor Physics and Device Processes

Introduction

Dr. Mohammadmahdi Ariannejad

Department of Electrical and electronic engineering

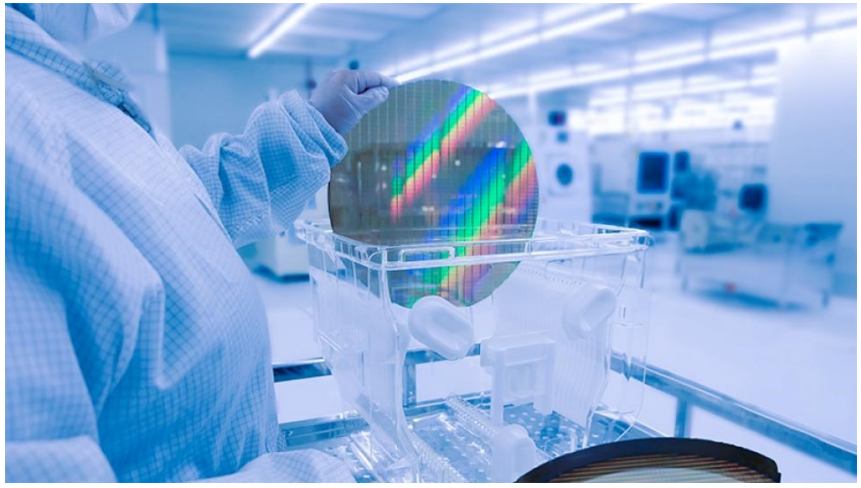
Office: A1-418

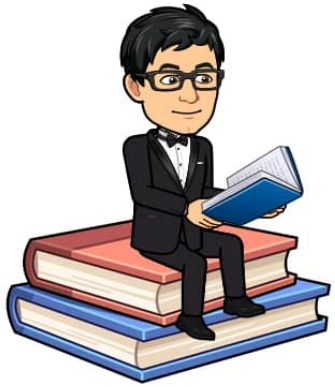
mohammadmahdi.ariannejad@xmu.edu.my

Lecture Outline

- **Semiconductors and the Integrated Circuit**
- Overview on the applications of semiconductors
- Fabrication of semiconductors

  Development of Integrated circuit (IC)

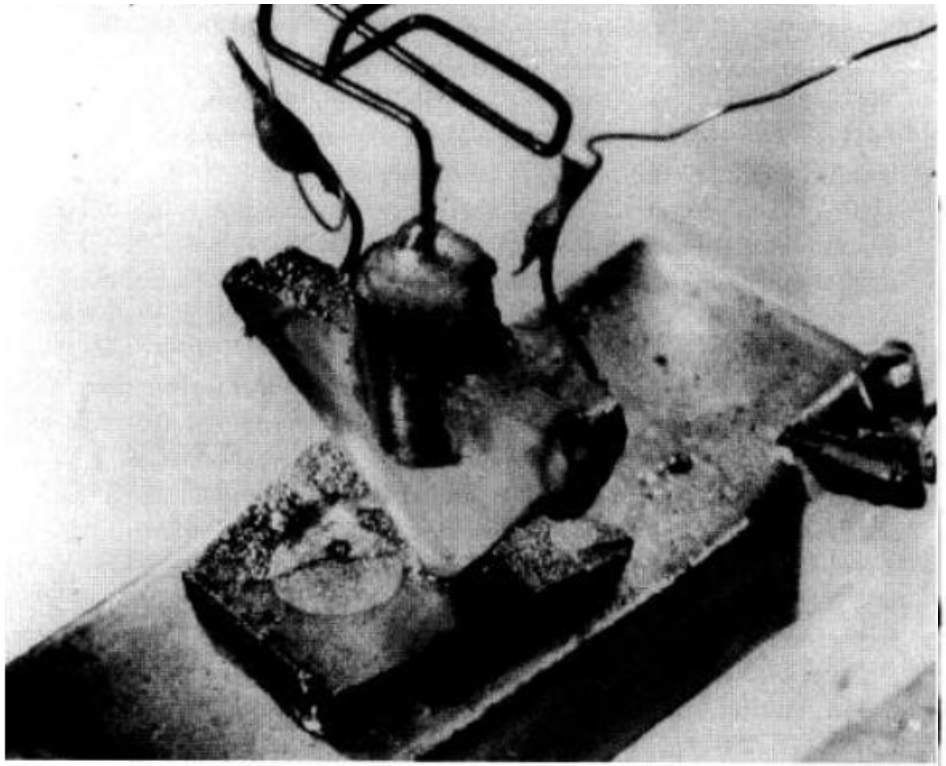




Semiconductors and the Integrated Circuit History and and Integrated Circuits (ICs)

- 1874, metal– semiconductor contact, Braun;
- 1942, thermionic-emission theory, Bethe
- World War II, Radar application, using highly pure Si/Ge, Ge diode
- 1947： Bell Telephone Laboratories , 1st transistor;
  - William Shockley, John Bardeen, and Walter Brattain, Nobel Prize(1956) in Physics
- 1958： Texas Instruments， IC/Si (1957, IC/Ge)
  - Kilby, Nobel Prize(2000) in Physics
- 1965: Moore's Law;
  - Intel, Gordon E. Moore, 1965
  - Alan Turing (1950), Douglas Engelbart(1960 lecture), New York Times, Aug.31,2009.

The first operating transistor



The first transistor.[7] (Photograph courtesy of Bell Laboratories.)



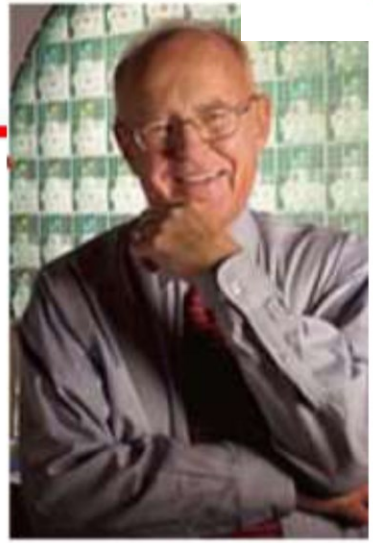
The first metal-oxide-semiconductor field-effect transistor.[13] (Photograph courtesy of laboratories.)

1947： Bell Telephone Laboratories , 1st transistor:

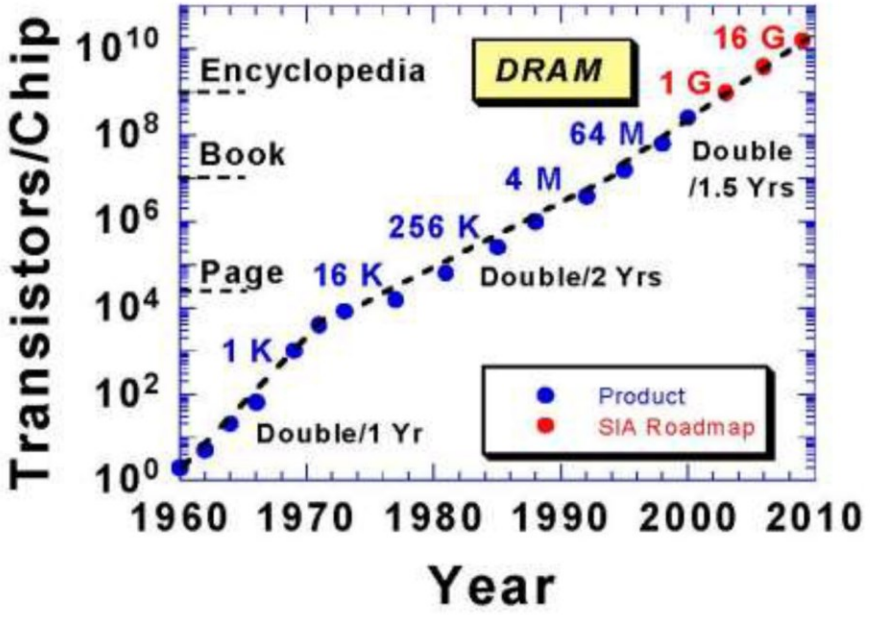
Nobel Prize(1956) in Physics: William Shockley, John Bardeen, and Walter Brattain

Moore's Law (1965)

- In 1965 Gordon Moore of Fairchild Semiconductor noted that the integrated circuit density had doubled in complexity each year since 1959 containing 50 transistors per chip (G.E. Moore, Cramming More Components onto Integrated Circuits, *Electronics* 38, 114-117, April 1965)
- He predicted that at that rate, there would be 6,500 components per chip in 1975. In 1975, Moore updated his prediction and proposed that density until that time had doubled every 2 years and would likely continue to do so (G.E. Moore, Progress in Digital Integrated Electronics, *IEEE IEDM Digest*, 11-13, Dec. 1975)



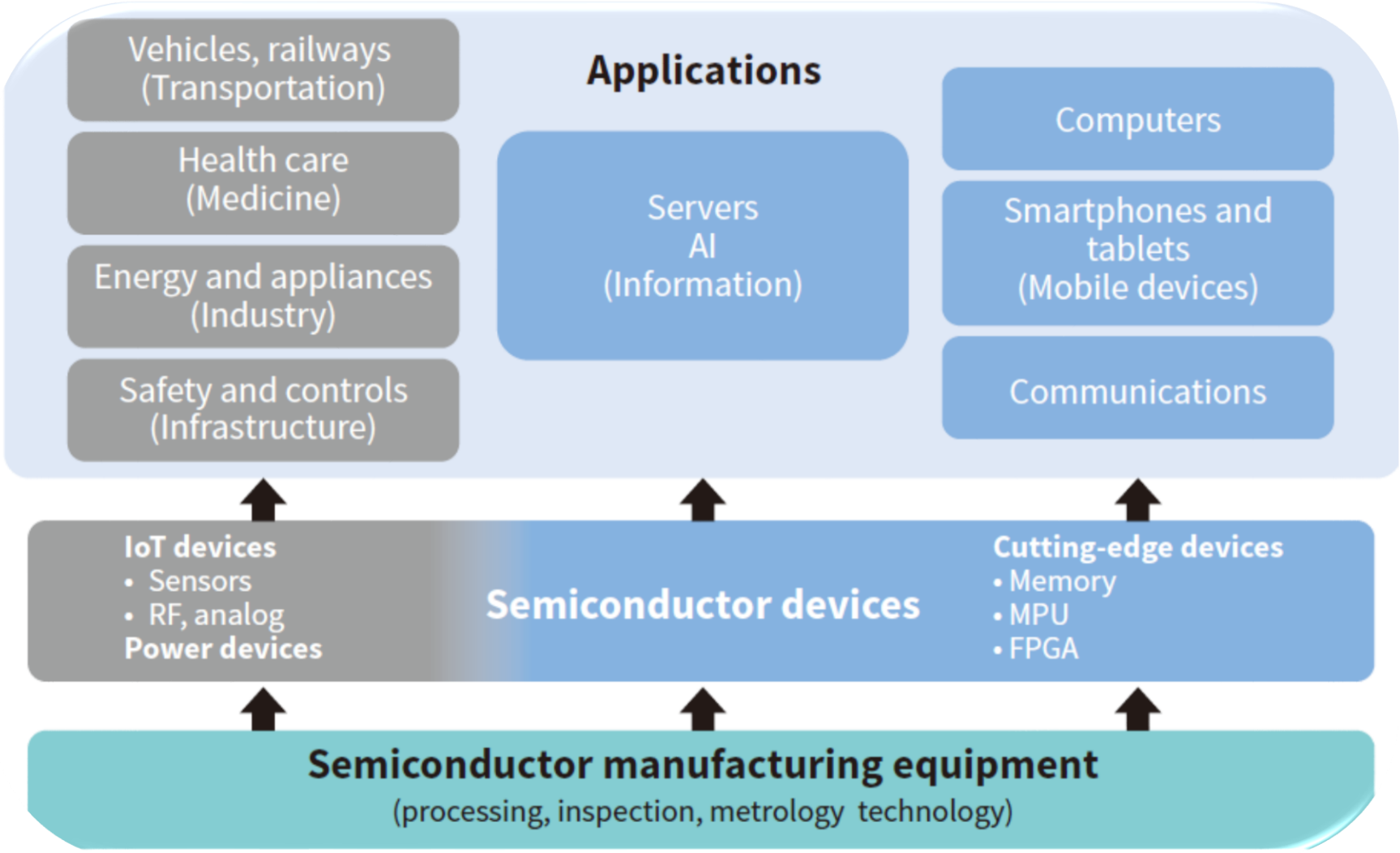
Gordon Moore



© D.K. Schroder, *Semiconductor Device Theory - 1*

Picture source: https://www.intel.com/pressroom/kits/events/moores_law_40th/

Overview on the applications of semiconductors



Scientific technologies driven by semiconductor device

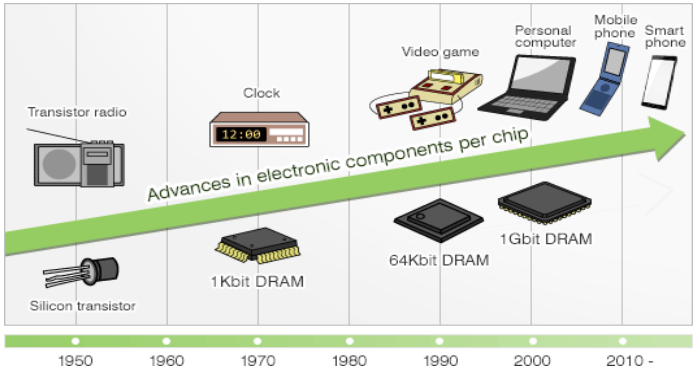

AUTOMOTIVE
COMMUNICATIONS
ROBOTICS
HEALTHCARE
ENERGY EFFICIENCY
AEROSPACE
INTERNET of THINGS
SAFETY & SECURITY
Integrated systems
Sub-system level devices: PCB, SMT
Packaged devices & modules
Components: wireless, photonic, electronic
Integrated Circuits / Chips
COMPOUND
SEMI-CONDUCTORS
SILICON
Internet of THINGS

Overview on the applications of semiconductors



Most semiconductor devices are fabricated on silicon wafer

Development of technology



Semiconductor chip





Qualcomm

HISILICON

Design companies

intel





Taiwan

Taiwan Semiconductor Manufacturing

South korea

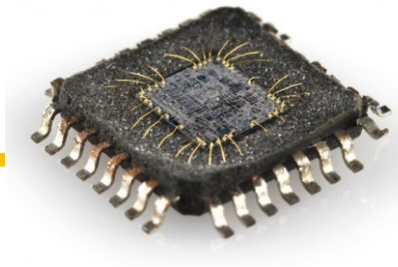




China

USA

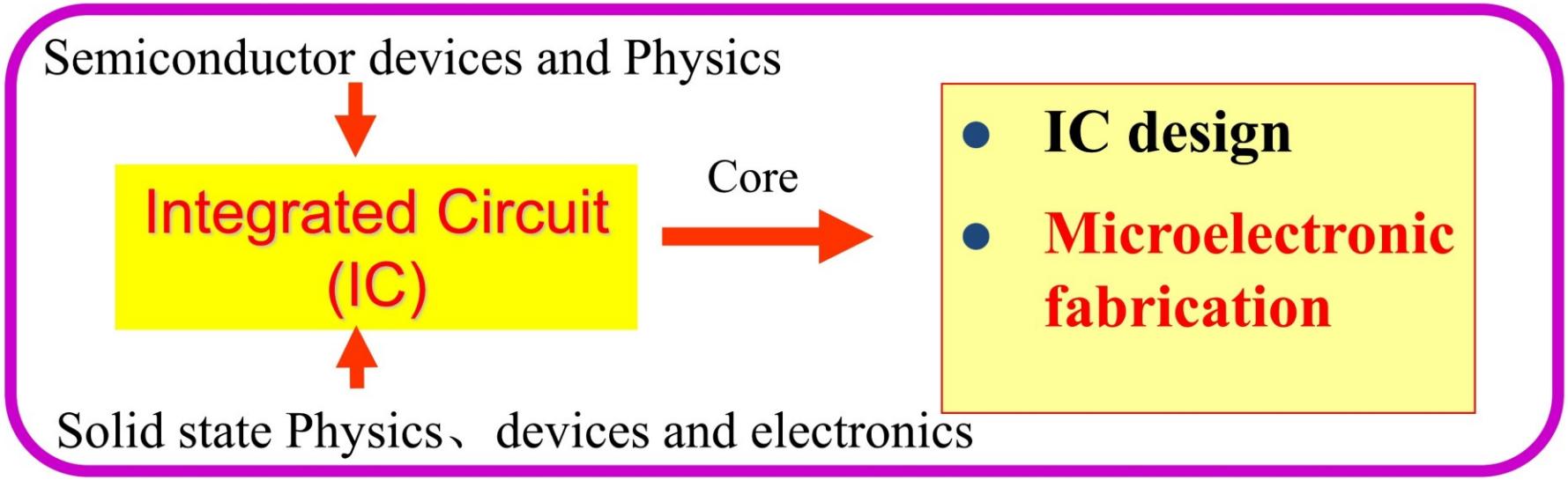
Fabrication



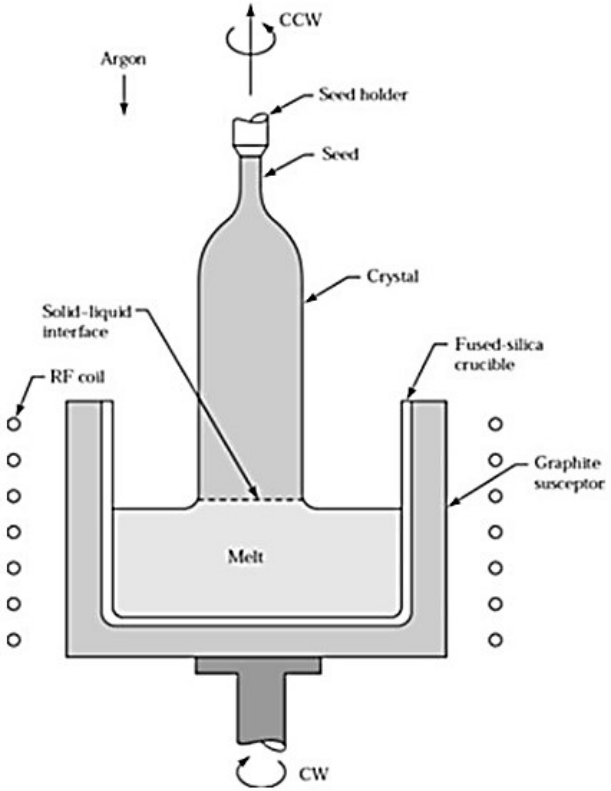
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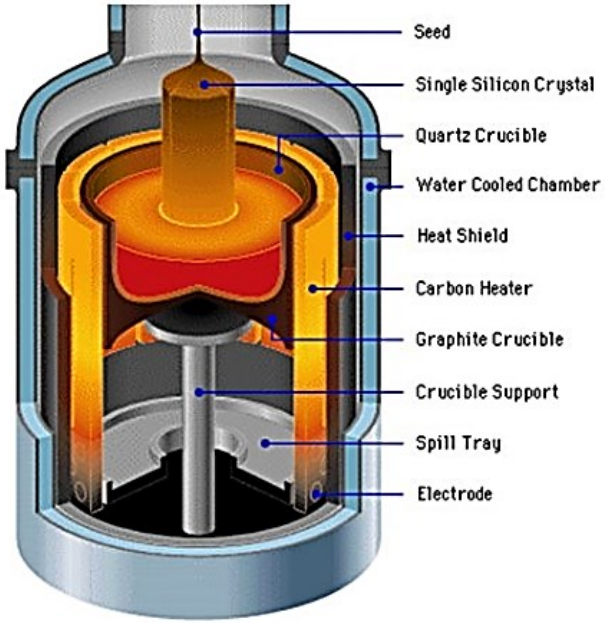
Integrated Circuit(IC):

- ✓ a set of **electronic circuits** on one small plate ("chip") of **semiconductor material**, normally **silicon**. ICs can be made very compact, having up to several billion **transistors** and other **electronic components** in an area the size of a fingernail.



Growth of semiconductor materials





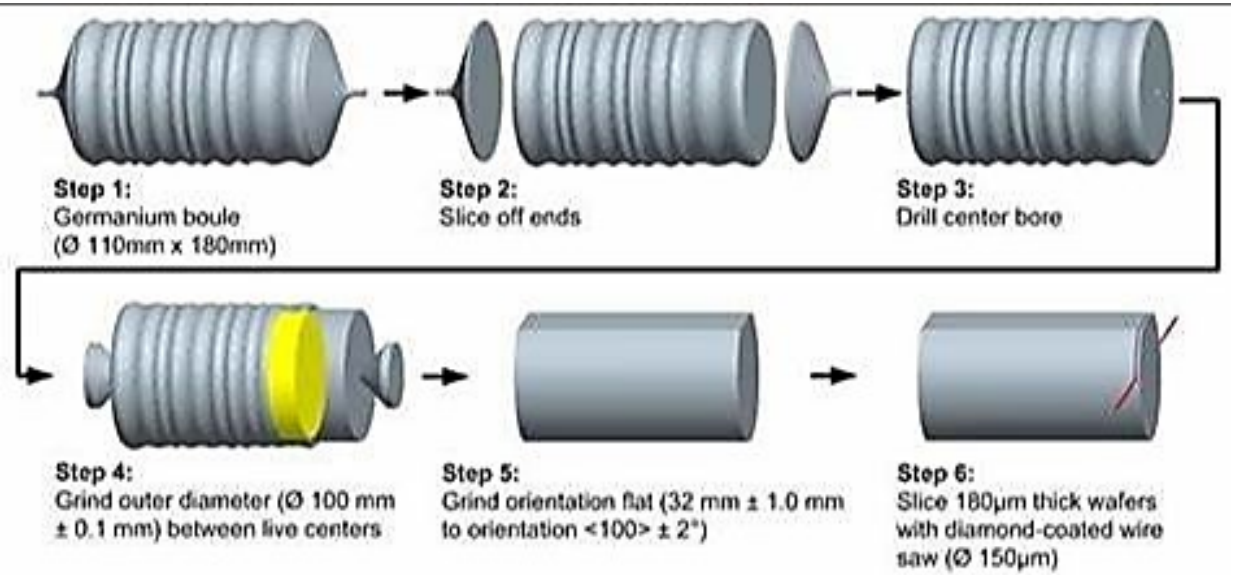
Making Silicon Crystal Wafers

A collection of short clips

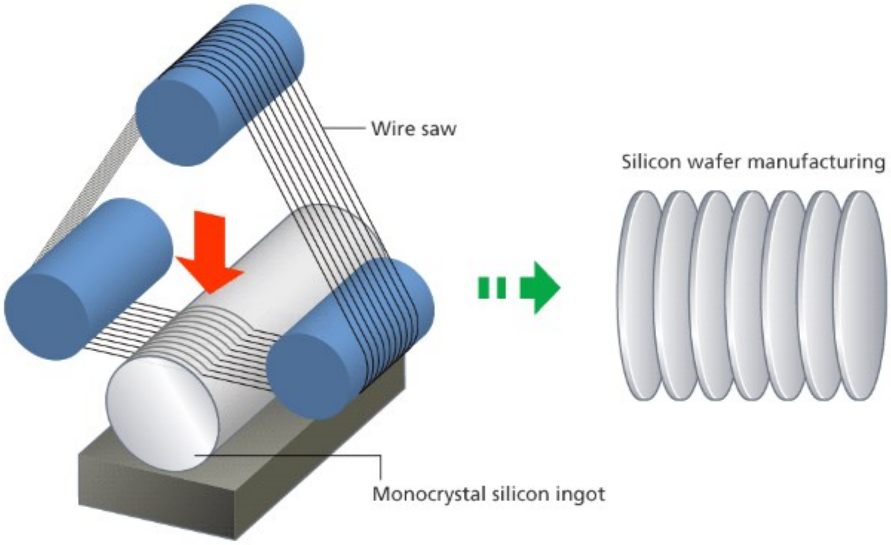
Liquid phase

Czochralski method

Single-crystal wafer manufacturing

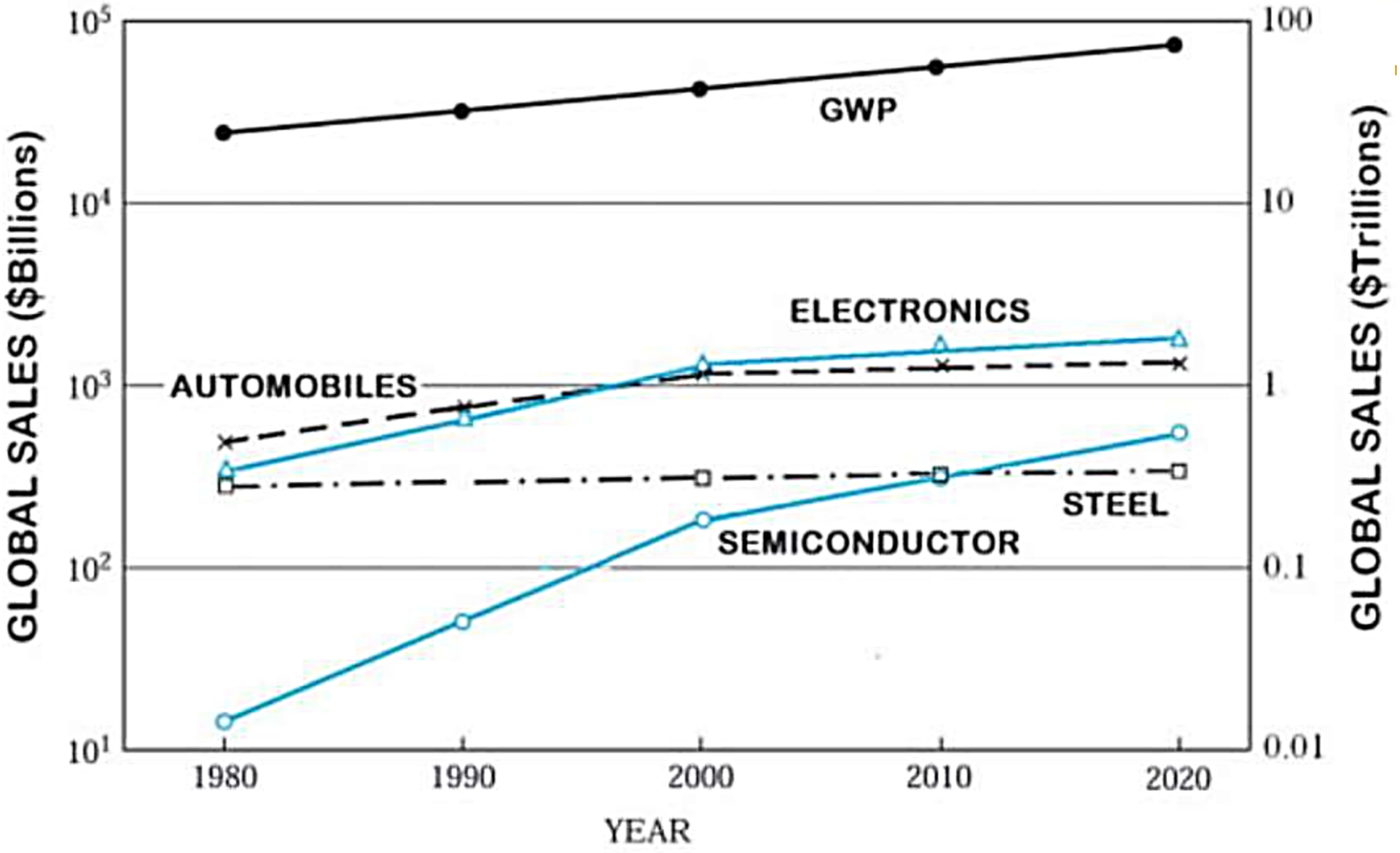


Ingot bar lapping



Slicing and lapping

https://companiesmarketcap.com/semiconductors/largest-semiconductor-companies-by-market-cap/



Gross world product (GWP) and sales volumes of the electronics, automobile, semiconductor, and steel industries from 1980 to 2010 and projected to 2020

1. *2009 Semiconductor Industry Report,* Ind. Technol. Res. Inst., Hsinchu, Taiwan, 2009; 2. Data from IC Insights, 2009.

Development of Integrated Circuit (IC)

Critical Dimension(CD) or Line width:

- **The smallest planar structure which can be fabricated by semiconductor processing. This is the important symbolic factor in electronic industry.**

Year	1999	2001	2003	2006	2009
Processing	0.18μm	0.15μm	0.13μm	0.1μm	0.07μm
Number of transistors	21M	40M	76M	200M	520M
Area/mm^2	340	385	430	520	620
Frequency/MHZ	1200	1400	1600	2000	2500
Number of metal layers	6~ 7	7	7	7~ 8	8~ 9
V_{dd}/V	1.65	1.35	1.35	1.05	0.75
Length/m	1480		2840	5140	10000
RAM/chip	25K		54K	230K	797K

IC Roadmap--Semiconductor Industry Association (SIA)

Semiconductors-conductors-insulators

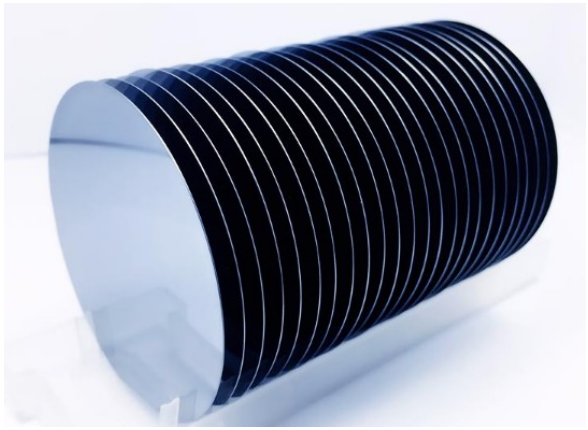
➢**Conductors:** material that easily conducts electrical current
The best conductors are single-element material: Al, Ag, Au, Cu
Only one valence electron very loosely bonded ---**free electron**

Insulators: material dose not conduct electrons
Valence electrons are tightly bound to the atom—**very few free electrons**

Semiconductors: material between conductor and insulator in its ability to conduct electric current
In its intrinsic state: neither a conductor nor an insulator



Copper

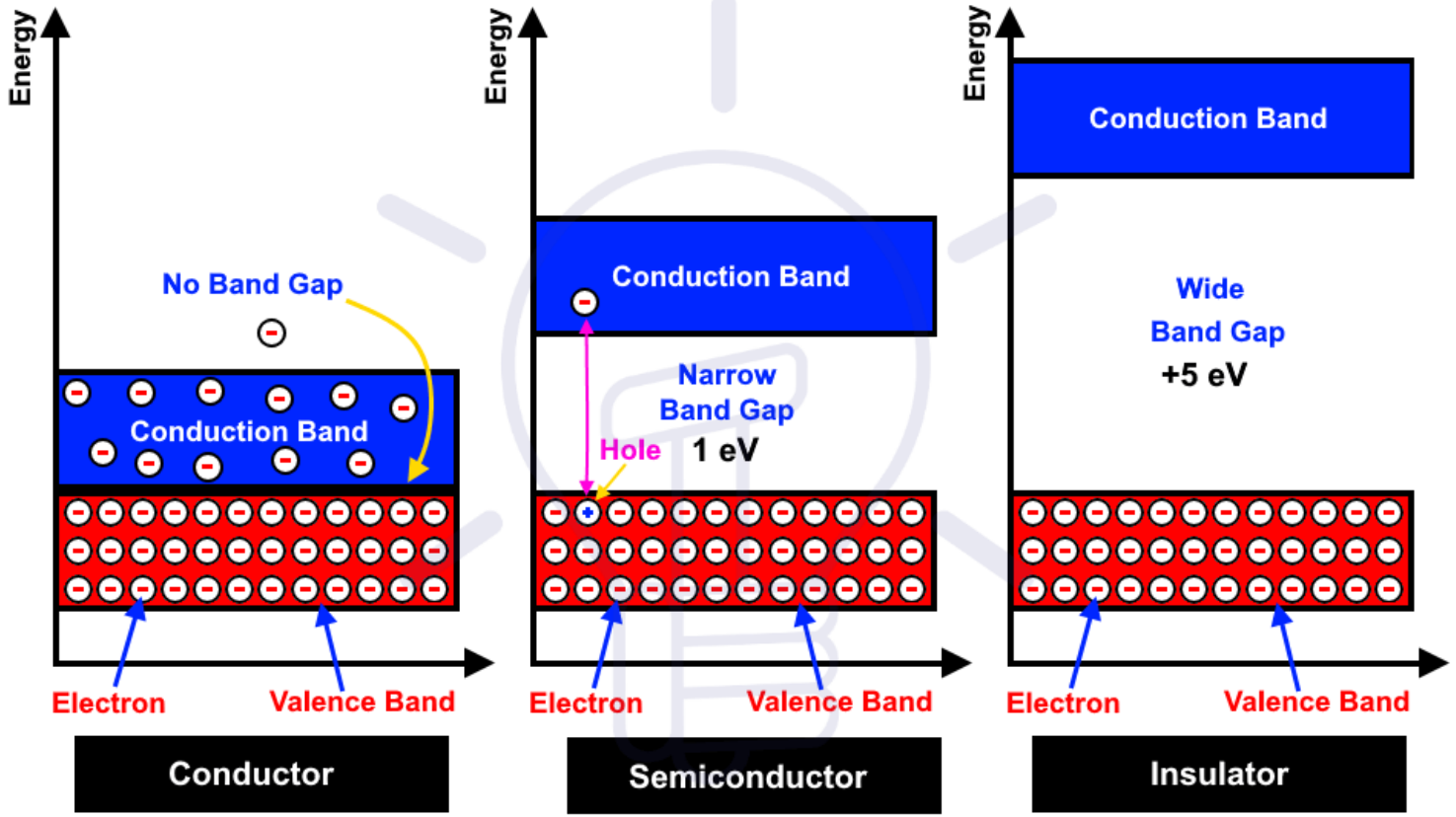


silicon



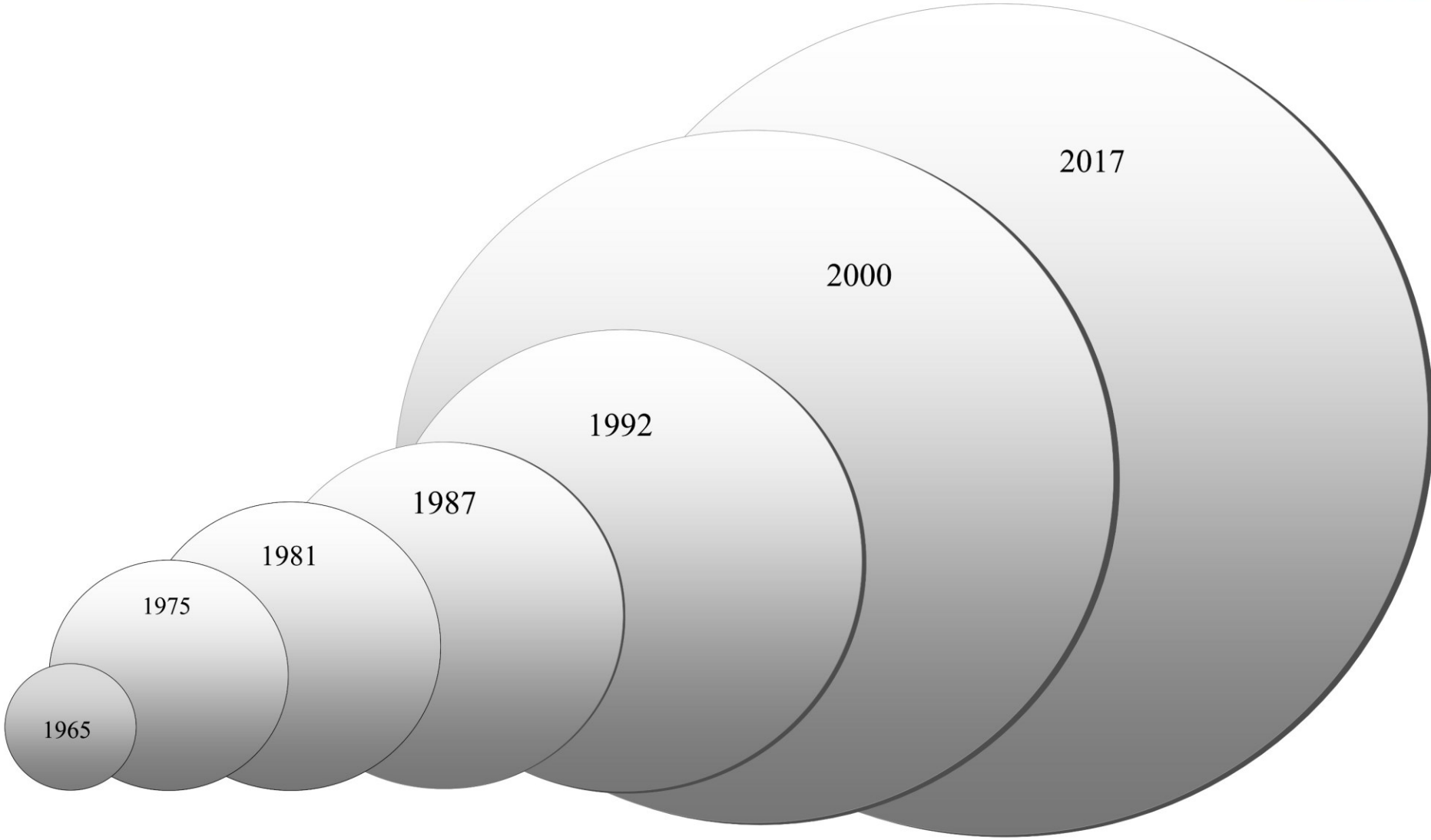
glass

Know the difference from Energy bands



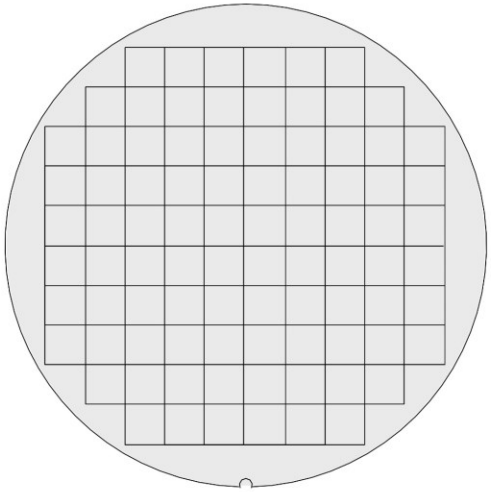
- Energy gap-the difference between the energy levels of any two orbital shells
- Band-another name for an orbital shell (valence shell=valence band)
- Conduction band –the band outside the valence shell where it has free electrons.

Evolution of Wafer Size



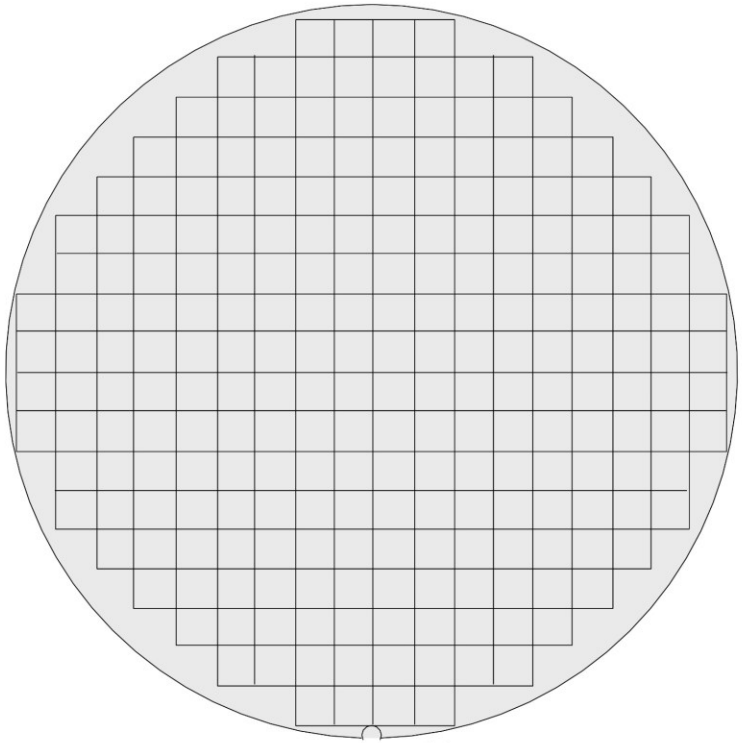
50 mm	100 mm	125 mm	150 mm	200 mm	300 mm	450 mm
2”	4”	5”	6”	8”	12”	18”

Increase in Number of Chips on Larger Wafer Diameters



88 die

200-mm wafer



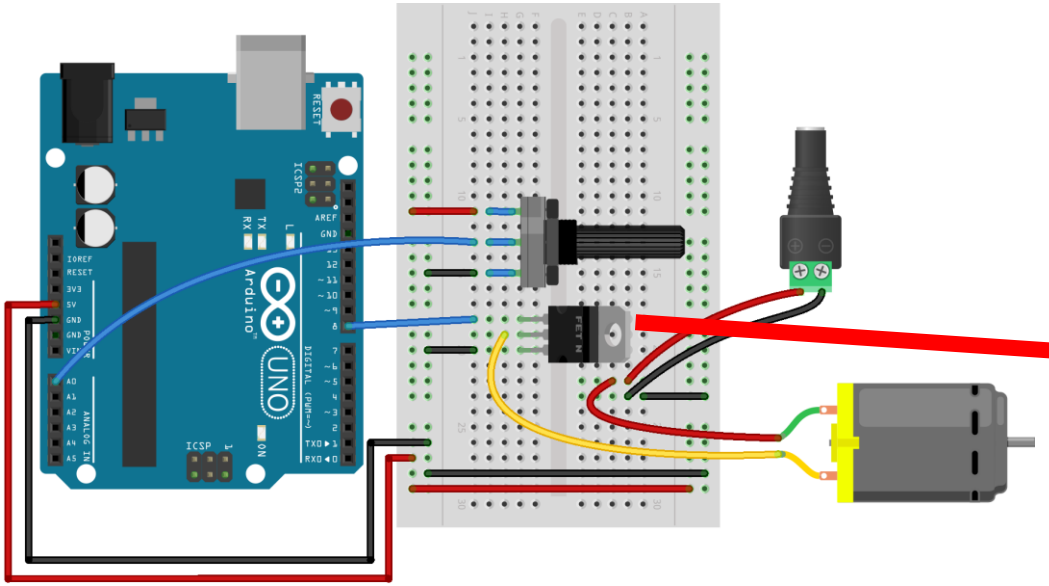
232 die

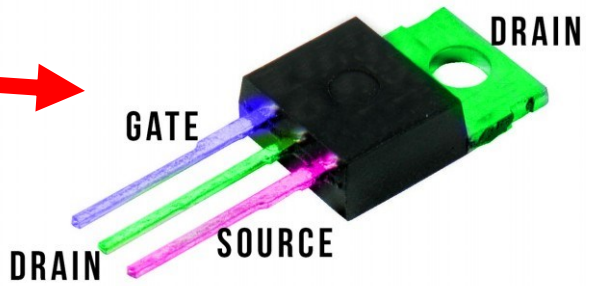
300-mm wafer

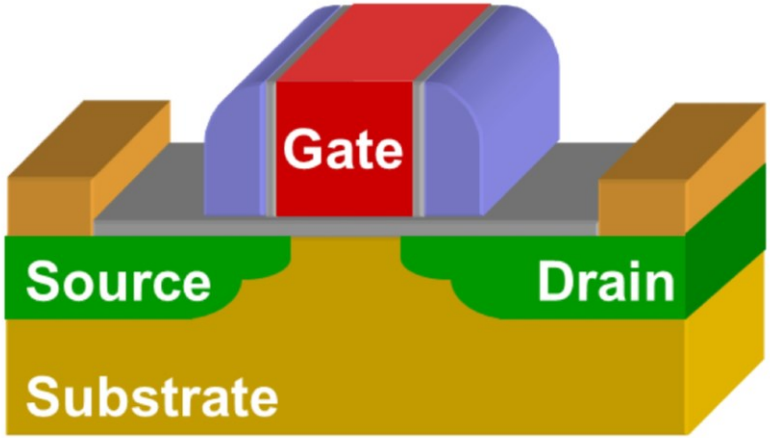
Assume large 1.5 x 1.5 cm microprocessors

Key semiconductor fabrication processing

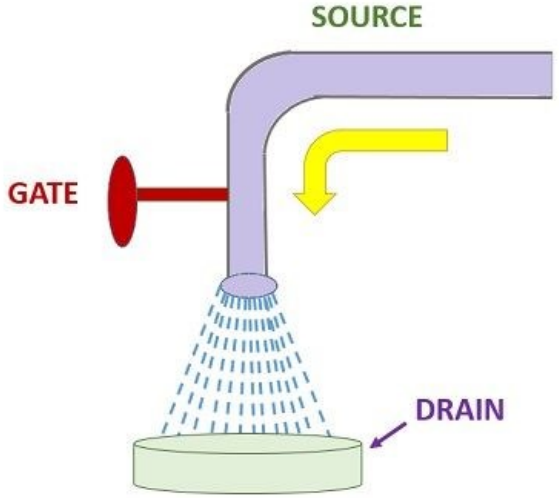
Year	Technology	Year	Technology
1918	Czochralski crystal growth	1963	CMOS
1925	Bridgman crystal growth	1967	DRAM
1952	III－V compounds	1969	Polysilicon self-aligned gate
1952	Diffusion	1969	MOCVD
1957	Lithographic photoresist	1971	Dry etching
1957	Oxide masking	1971	Molecular beam epitaxy
1957	Epitaxial CVD growth	1971	Microprocessor
1958	Ion implantation	1982	Trench isolation
1958	Hybrid integrated circuit	1989	Chemical mechanical polishing
1959	Monolithic integrated circuit	1993	Copper interconnect
1960	Planar process		







Metal-Oxide-Semiconductor (MOS)
Field-Effect Transistor (FET)



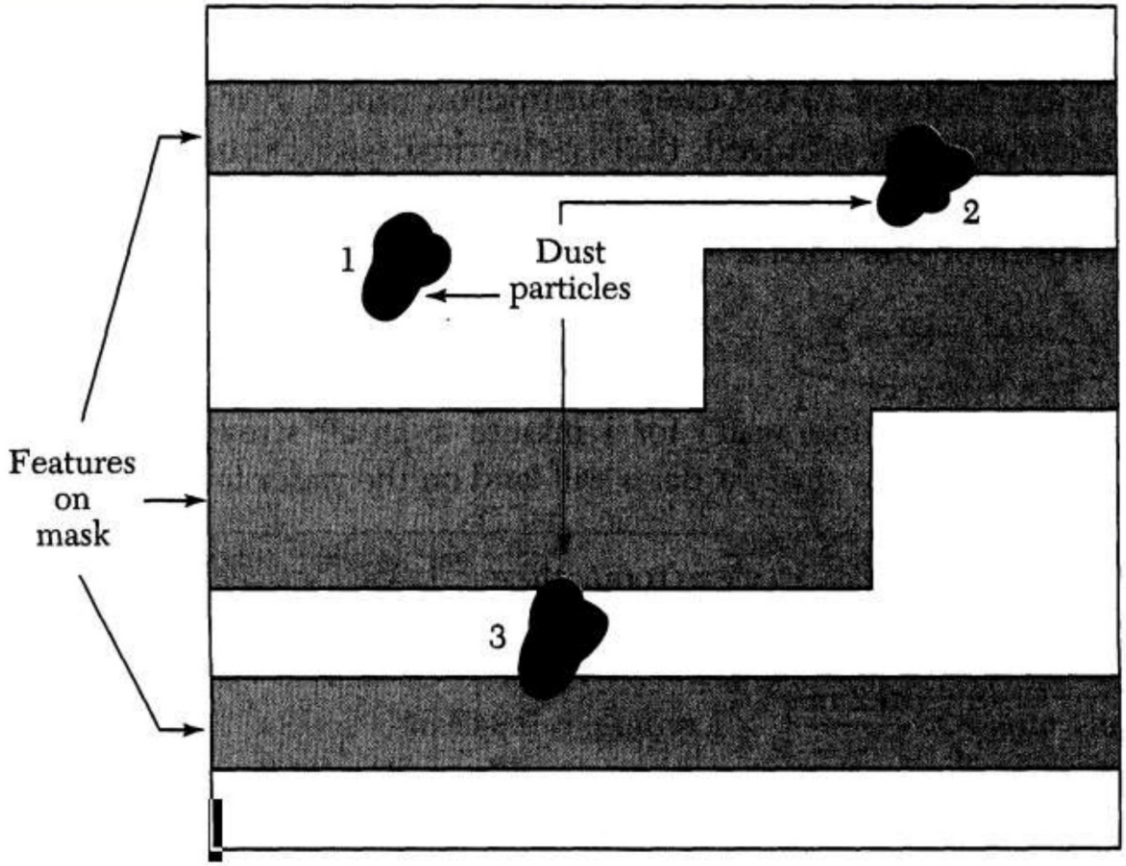
Water Analogy for Understanding FET



clean room (class 1000)

The Clean Room

- A **cleanroom** is an environment, typically used in manufacturing or scientific research, that has a low level of environmental pollutants such as dust, airborne microbes, aerosol particles and chemical vapors.
- An IC fabrication facility requires a clean room ,especially in the area used for lithography .



lithography machine



End of lecture Introduction! ☺



- Additional readings:
  - **Physics and Properties of Semiconductors-A Review**

    (Chapter1, S. M. Sze, M.-K. Lee Semiconductor Devices: Physics and Technology, 3rd Edition, Wiley 2012)

- Next lecture: **The Crystal Structure of Solids**

- Questions?



